

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: i am looking for connector applications for uth

Assistant: To better understand your requirements for connector applications under the hood (UTH):

1. What is the exact use of the polymer in your connector application? For example, is it for the connector housing, sealing element, terminal retainer, or another component?
2. Where will this connector be used within the vehicle's under-hood environment? For instance, is it part of the engine wiring harness, sensor interface, power distribution, or another system?

Your answers will help me identify the most suitable material options for your application.

User: Yes it is for the housing. I need it to be resistant to oil and fuel,It is for a sensor connector

Assistant: Thank you for clarifying. To proceed, could you please provide:

1. What is the minimum wall thickness of your sensor connector housing? (This helps ensure proper material flow and mechanical strength.)
2. Do you have any preferred base materials for the housing, such as specific polymers you've used before (e.g., PBT, PA, PPS, etc.)?
3. Are you open to alternative material suggestions, or do you require the same base

material as previously used?

Also, to confirm: The sensor connector housing will be used under the hood, exposed to oil and fuel, and must withstand high temperatures, vibration, and thermal cycling. If any of these assumptions are incorrect, please let me know.

User: I am open to suggestions on material, no preference on grades

Assistant: Here are the key material requirements (CTQs) assumed for your sensor connector housing application under the hood:

1. Injection molding processing
2. Oil and fuel resistance
3. Flame retardant (V-0 rating at 0.8 mm minimum wall thickness)
4. Heat stabilized for long-term durability
5. Electrical insulation
6. Dimensional stability
7. Thermal shock resistance

Are all of these requirements correct for your application? Please confirm if these match your needs, or let me know if any should be changed.

User: i dont need ul property and i want upper quartile of charpy impact strength

Assistant: Thank you for your feedback. To ensure the material selection matches your needs, could you please confirm or provide values for the following:

1. What is the target value or minimum requirement for Charpy impact strength? If you want the upper quartile, do you have a reference value or range in mind?
2. Is glass fiber reinforcement required for your connector housing, and if so, what

percentage (e.g., 15–35%)?

3. Is hydrolysis resistance important for your application, considering exposure to moisture under the hood?

Once I have your input on these, I can proceed with grade suggestions that best fit your requirements.

User: i dont have that information recommend grades based on the ctq assumed so far

Assistant: Based on your requirements for a sensor connector housing under the hood—resistant to oil and fuel, heat stabilized, electrically insulating, dimensionally stable, hydrolysis resistant, and targeting upper quartile Charpy impact strength—the following grades may be suitable options:

1. Zytel® HTN54G35HSLR NC010 (PPA, 35% GF, toughened): Excellent chemical and thermal robustness, high impact strength, and proven performance in under-the-hood sensor connector housings.
2. Zytel® HTN51G35EF NC010 (PPA, 35% GF): Benchmark PPA for connector housings, strong dimensional stability, and good impact performance.
3. Zytel® 80G33HS1L NC010 (PA66, 33% GF, super tough): Very high impact strength, widely used for under-the-hood housings, good oil resistance.
4. Zytel® 70G35HSL NC010 (PA66, 35% GF): Robust, cost-effective solution for under-the-hood housings, strong impact and stiffness.
5. Zytel® 70G30HSLR BK186LM (PA66, 30% GF, hydrolysis resistant): Balanced option with hydrolysis resistance and adequate impact strength; black variant included as no natural variant is available for this formulation.

These grades are potential options based on technical data and typical application

requirements. Final material selection and product design decisions should be made based on your specific performance targets and testing. Please ensure compliance with all applicable standards and regulations for your application. If you need further customization or have more specific requirements, let me know!

*Note: This recommendation is generated by an AI assistant and should be considered purely informational.*